UAV Online Trajectory Planning Using Confidence Bounds of Chance-Constrained Geo-Fences

***Abstract*—This project report is based on my current research in UAV online trajectory planning. The goal of this project is to develop a learning-based algorithm that is devised to build the upper confidence bound of the unknown geo-fences in real-time. This algorithm will assist a group of UAVs to converge together and re-formation from random geo-position in an online manner with geo-fence avoidance.**

I. Introduction and Motivation

The last few decades have witnessed a tremendous surge of research interest in applications of autonomous vehicles, especially the unmanned aerial vehicles (UAVs), such as transportation systems [1]–[3], package delivery [4]–[6], environmental monitoring [7], and etc. To achieve the desired autonomy, planning an optimal trajectory while ensuring safety, i.e., avoiding the potential obstacles and collisions during the flight, has been one of the central technical problems in operating UAVs. As a natural consequence, plenty of methodologies have been developed in the literature for solving the problem of UAV trajectory planning.

There are mainly two categories for the existing path planning approaches: i) global planners, which are often carried out before the execution of the flight. i.e., in an offline manner; and ii) local planners, which generate the trajectory incrementally during the flight, i.e., in an online manner. The typical global planners include graph search methods (e.g., Dijkstra's algorithm [8] see Fig. 1, A* algorithm [9]),

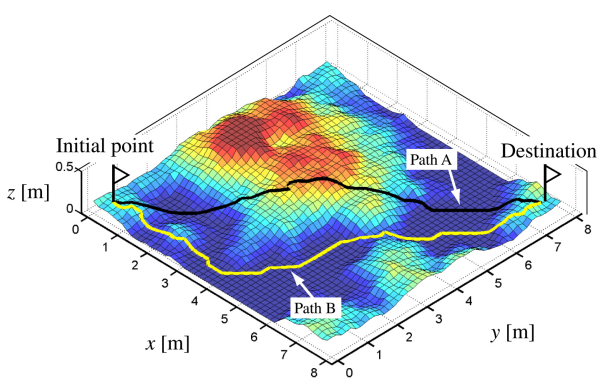

Fig. 1. Graph-Based Path Planning: Dijkstra's Algorithm.

sampling based methods, e.g., RRT/RRT* algorithm [10], [11] see Fig. 2, and mixed-integer linear programming (MILP) methods [12]. While the above global planners contains the

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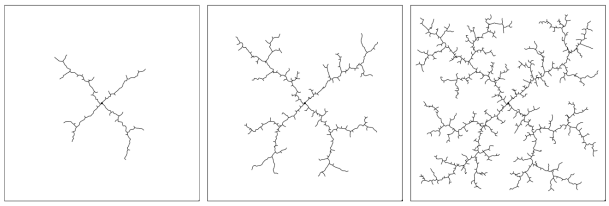

Fig. 2. Incremental build of a rapidly exploring random tree (RRT). — Credit: S. LaValle

feature that the performance guarantee to provide a globally optimal trajectory, such a process is usually time-consuming and heavily relies on the knowledge of environment as well as the static obstacles is completely known. To cope with the issue state above, various local planner have been proposed in the literature. For example, the idea based on the notion of artificial potential field, which plans the UAV trajectory in an online manner, by tracking along the negative gradient direction of the potential field. The method of model predictive control (MPC) [13], which solves an optimal control problem at each time-step and executes the first step in the resulting optimal controller. More recently, perception-aware trajectory planning algorithms [14], [15] are proposed, which the environment perception data is collected by the on-board sensors and then used to update the trajectory during the flight. Based on the methods mentioned above, this project will aim to develop an online UAV trajectory planning algorithm which integrates the perception of a partially unknown environment and real-time refinement of the trajectory planning.

Furthermore, the uncertainty of the environment and/or the motion of UAVs are also a key component for UAV trajectory planning. To deal with the potential uncertainties, multitudinous methods have been proposed in the literature modeled on various techniques, including set-bounded uncertainty model [16], robust optimization [17], [18], chance-constrained optimization [19]–[21], etc. Whereas the first two are essentially deterministic methods which render robustness of the trajectories by ensuring the performance under the worst scenario. The chance-constrained optimization approach is usually able to produce less conservative trajectories in probabilistic manner while the safety is guaranteed. More specifically, the authors in [22] took advantages of the assumption on a linear Gaussian system, the chance constraints in [22] are approximated by deterministic ones relying on the techniques presented in [23]. In this project, the planning procedure is particularly formulated as an information-cost stochastic nonlinear optimal control problem, where the safety is ensured

by chance constraints and the objective is to maximize the exploration gain and minimize control effort simultaneously for several quad-rotor UAVs to converge together from random locations in the global map. By using statistical moments of the uncertainty distribution, the risk of collision is bounded by concentration inequalities. On this account, by leveraging Gaussian distributions to deal with the chance constraints, the trajectory planner could address a broad range of probability distributions involved in the uncertainty of agent's dynamics [24].

It is important to mention that the idea from [25], a novel UAV trajectory planning algorithm following the technique of chance-constrained optimization, proposed that each geo-fence can be modeled as a convex polygon, but its boundaries are subject to some uncertainties, i.e., the perpendicular distance of each boundary to the center of the polygon is uncertain. By assuming that the uncertainties follow known probability distributions (not necessarily Gaussian), a sampling based solution method is then presented to solve the formulated chance-constrained optimization. However, several concerns regarding to the proposed method are still critical and remain unsolved. First, the planner presented in [25] is inherently offline. i.e., the global environment is pre-informed, all samples of the uncertain distances are required to be obtained beforehand. Second, to ensure the safty, more sample would be preferred, which would significantly increase the time complexity of solving the optimization problem, which prevents it turn into a local planner solution. Third, the assumption on cumulative distribution function is restrictive and it will also further increase the time complexity of the proposed method.

Therefore, the objective of this project is to develop a new local trajectory planner, inspired by [25], no longer relies on the sampling of the uncertain variables but builds on the real-time perception of the unknown geo-fences. To do so, the UAV is assumed to be capable of receiving real-time measurements, possibly with noise from sensors equipped on-board on the surrounding environment. To further ensure the UAV's safety, the *dummy upper confidence bound* (D-UCB), which was firstly proposed in [26], is used to construct a set of appropriate bounds for the unknown distances. It should be remarked that the D-UCB concept was firstly proposed in [26] and has significant differences against the standard upper confidence bound (UCB) which has been widely studied in the context of machine learning [27]–[29]. Due to the special structure of the D-UCB, we note that the constructed bounds are exceptionally suitable to be applied in solving the chance-constrained optimization. We leave the details to be discussed in Section IV-B.

The rest of this paper is organized as follows. Section II first presents the basic formulation of the UAV trajectory planning problem. Section III-A develops an online estimator for the unknown environmental information. Section III-B then builds the key notion of D-UCB based on the obtained estimation. The complete UAV online trajectory planner is given in Section IV-A and its performance, including both feasibility and optimally, is analyzed in Section IV-B. Since this research is still in progress, Section V will provide an general idea of my plan of the simulation, result analysis, and the potential future improvements.

II. Problem Formulation

As mentioned earlier, this project is based on the previous trajectory planning problem as proposed in [25], which is specified as the chance-constrained quadratic programming (CC-QP) problem. The necessary notation as well as the CC-QP formulation will also be introduce briefly in this section.

A. *The Basic CC-QP Formulation*

Suppose that the UAVs are located in a 2-D plane, is denoted as $\mathbf{x}(t) \in \mathbb{R}^2$ at time-step $t \in \mathcal{T} := \{0, 1, ..., T\}$ where T represents a finite planning horizon. Furthermore, we denote the velocity and control input of the UAV as $\mathbf{V}(t) \in \mathbb{R}^2$ and $\mathbf{u}(t) \in \mathbb{R}^2$, respectively. Then, the CC-QP formulation of the UAV trajectory planning problem is expressed as follows:

$$\min_{\{\mathbf{u}(t)\}_{t\in\mathcal{T}}} \sum_{t=0}^{T-1} ||\mathbf{u}(t)||^2 \tag{CC-QP}$$

$$s.\ t. \begin{bmatrix}\mathbf{x}(t+1)\\ \mathbf{v}(t+1)\end{bmatrix} = A\begin{bmatrix}\mathbf{x}(t)\\ \mathbf{v}(t)\end{bmatrix} + B\mathbf{u}(t), 0 \le t \le T-1; \tag{UAV Dynamics}$$

$$\mathbf{x}(0) = \mathbf{x}^{orig}, \mathbf{x}(T) = \mathbf{x}^{dest}; \tag{Origin-Destination Pair}$$

$$\mathbf{x}(t) \in \mathcal{K}, t \in \mathcal{T}; \tag{Bounded Planning Space}$$

$$\mathbf{Pr}\big(\mathbf{x}(t) \notin \mathcal{S}_i^{\mathbf{r}}, \forall i \in \mathcal{I}\big) \ge \alpha, t \in \mathcal{T} \tag{Geo-Fence Restriction}$$

Note that, in the above formulation, the objective is to minimize the total control effort during the planning horizon; $A \in \mathbb{R}^{4\times 2}$ and $B \in \mathbb{R}^{4\times 2}$ denote the system matrices of the UAV's linear dynamics; $\mathbf{x}^{orig} \in \mathbb{R}^2$ and $\mathbf{x}^{dest} \in \mathbb{R}^2$ assign a pair of origin and destination positions for the UAV; and $\mathcal{K} \subset \mathbb{R}^2$ defines a bounded planning space which is supposed to be convex and usually specified by a set of linear inequalities, i.e., $\mathcal{K} := \{\mathbf{x}|L\mathbf{x} \le \mathbf{b}\}$. More importantly, the last constraint, representing the chance-constrained geo-fence restriction, ensures that the probability of a collision to any restricted area, defined by $\mathcal{S}_i^{\mathbf{r}}, i \in \mathcal{I} := \{1, 2, ..., I\}$, has to be less than a pre-determined threshold . Note that the probability $\mathbf{Pr}(.)$ is taken with respect to the uncertain variable r which is related to the shapes of the restricted areas; see details later. In this paper, we consider that there are totally *I* restricted areas in the planning space $\mathcal{K}$ and each of them corresponds to a geo-fence which will be introduced in the next subsection.

B. *Chance-Constrained Geo-Fence Restriction*

Similar to the model which has been applied in [25], each of the geo-fences by a convex polygon $\mathcal{S}_i^{\mathbf{r}}$, as exemplified in Fig. 3 Here, a fixed point $\mathbf{P}_i \in \mathbb{R}^2$ is used to denote the center

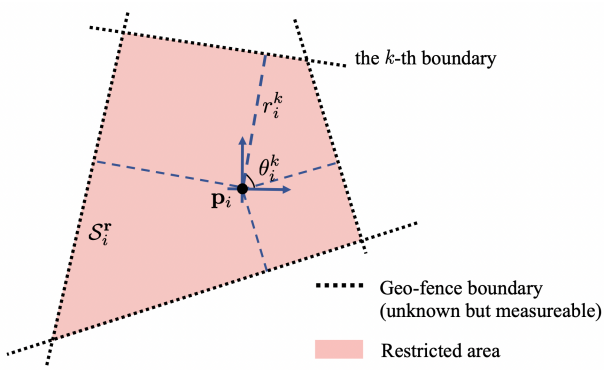

Fig. 3. Illustration of the convex geo-fence $\mathcal{S}_i^{\mathbf{r}}$

of the polygon $\mathcal{S}_i^{\mathbf{r}}$, and a pair of the angle and perpendicular distance, i.e., (θ_i^k, r_i^k), is used to represent each boundary of $\mathcal{S}_i^{\mathbf{r}}$, indexed by $k \in \{1, 2, ..., K_i\}$.. More specifically, all boundaries of the polygon can be then analytically expressed as,

$$\begin{bmatrix}\cos(\theta_i^k)\\ \sin(\theta_i^k)\end{bmatrix}^\top (\mathbf{x} - \mathbf{p}_i) - r_i^k = 0, \forall k \in \{1, 2, ..., K_i\}. \quad (1)$$

As a consequence, the restricted area defined by the geo-fence $\mathcal{S}_i^{\mathbf{r}}$ with K_i boundaries is,

$$\mathcal{S}_i^{\mathbf{r}} := \left\{\mathbf{x} \middle| \begin{bmatrix}\cos(\theta_i^k)\\ \sin(\theta_i^k)\end{bmatrix}^\top (\mathbf{x} - \mathbf{p}_i) < r_i^k, \right. \\ \left. \forall k \in \{1, 2, ..., K_i\}, \mathbf{x} \in \mathbb{R}^2\right\}. \quad (2)$$

Remark 1: In fact, the above model of the geo-fences can be easily extended to a 3-D case, in which the "boundary" of the geo-fence will become a plane of the polytope. Suppose that pi also denotes a fixed point which locates inside the polytope, then the restricted area defined by the 3-D geo-fences can be expressed as

$$\mathcal{S}_i := \left\{\mathbf{x} \middle| \begin{bmatrix}\sin(\theta_i^k)\cos(\phi_i^k)\\ \sin(\theta_i^k)\sin(\phi_i^k)\\ \cos(\theta_i^k)\end{bmatrix}^\top (\mathbf{x} - \mathbf{p}_i) < r_i^k, \right. \\ \left. \forall k \in \{1, 2, ..., K_i\}, \mathbf{x} \in \mathbb{R}^2\right\}. \quad (3)$$

where r_i^k still represents the perpendicular distance from the point $\mathbf{P}_i$ to the k-th plane (denote the corresponding point on the plane as $\mathbf{q}_i^k$); θ_i^k and ϕ_i^k are the polar and azimuth angles of the point $\mathbf{q}_i$, respectively.

To guarantee the UAV's safety, the geo-fence restriction in the CC-QP formulation enforces the UAV (with probability no less that α) to bypass all restricted areas at any time-step t, i.e., $\mathbf{x}(t) \notin \mathcal{S}_i^{\mathbf{r}}, \forall i \in \mathcal{I}, t \in \mathcal{T}$. Though the geo-fences are assumed to be convex, yet the feasible region for the UAV with respect to each $\mathcal{S}_i^{\mathbf{r}}$, denoted as $\mathcal{C}_i^{\mathbf{r}} := \mathcal{K}/\mathcal{S}_i^{\mathbf{r}}$, is not necessarily convex. Therefore, to tackle with the non-convexity, a group of binary variables $z_i^k \in \{0, 1\}$ is further introduced, such that the feasible region $\mathcal{C}_i^{\mathbf{r}}$ can be equivalently expressed as

$$\mathcal{C}_i^{\mathbf{r}} := \left\{\mathbf{x} \middle| \begin{bmatrix}\cos(\theta_i^k)\\ \sin(\theta_i^k)\end{bmatrix}^\top (\mathbf{x} - \mathbf{p}_i) + M(1 - z_i^k) \geq r_i^k, \right. \\ \left. \sum_{k=1}^{K_i} \geq 1, z_i^k \in \{0, 1\}, \mathbf{x} \in \mathcal{K}\right\}. \quad (4)$$

where M is a constant which is supposed to be large enough. Now, with the previous geo-fence restriction replaced by the following form, in which the binary decision variables z_i^k's are involved,

$$\mathbf{Pr}(\mathbf{x}(t) \in \mathcal{C}_i^{\mathbf{r}}, \forall i \in \mathcal{I}) \geq \alpha, \forall t \in \mathcal{T}, \quad (5)$$

the CC-QP formulation has actually been turned to a chance-constrained mixed-integer quadratic program (CC-MIQP).

In contrast of what the authors assumed in [25] for the perpendicular distance $\mathbf{r}_i^k$'s, which are corresponding to a set of random variables, following some known probability distributions. Here, a different scenario is considerd where the distances $\mathbf{r}_i^k$'s are completely unknown and expect to develop an online trajectory planner. That is, the unknown distances $\mathbf{r}_i^k$'s are measured with noise by some on-board sensors and the trajectory is updated in real-time based on an appropriate surrogate of $\mathbf{r}_i^k$'s during the flight. It is important to remark that, since the obtained measurements are subject to random noises and the Bayesian type estimator will be developed to handle it, with slight abuse of notation, the probability $Pr()$ involved in the chance constraint (5) as well as in the previous CC-QP formulation is still explainable in the following discussions.

C. *Consensus of Destination Location for Multi-Agent*

Recall the origin-destination pair, $\mathbf{x}(0) = \mathbf{x}^{orig}$, and $\mathbf{x}(T) = \mathbf{x}^{dest}$, for Multiple UAVs, denote each has $\mathbf{x}_i^{orig}$ at $\mathbf{x}(0)$, and $\mathbf{x}_i^{dest}$ at $\mathbf{x}(T)$. Assume the destination is located at the geometric center of these UAVs' $\mathbf{x}_i^{orig}$, then the rendezvous destination can be written as:

$$\mathbf{x}^{dest} = \frac{1}{I}\sum_{i=1}^{I} \mathbf{x}_i^{orig} \quad (6)$$

where $i \in \{1, 2, ..., I\}$ and I is the total number of agents. However, in 2D plane, it is impossible for all agent rendezvous to the exact same point. To avoid collision, the final destination for all members can be a range within certain radius, denoted as $\mathcal{D}$, where

$$\mathcal{D} := \left\{\mathbf{x} \mid ||\mathbf{x} - \mathbf{x}^{dest}||_2 \leq b\right\} \quad (7)$$

where b is the radius at the destination of the re-formation.

III. CONFIDENCE BOUND OF THE DISTANCES

In this section, let us first develop a recursive least squares (RLS) method for estimating the unknown r_i^k's, upon which confidence bounds are then constructed to provide the surrogate of r_i^k's.

A. *RLS Estimator*

For the sake of presentation, let us first stack r_i^k's for all geo fences as a compact vector $\mathbf{r}_0 \in \mathbb{R}_+^K$ with $K = \sum_{i=1}^{I} K_i$, i.e., $\mathbf{r}_0 = [r_1^1, r_1^2, ..., r_1^{K_1}, ..., r_I^1, r_I^2, ..., r_I^{K_I}]^\top$. Next, we consider the following stochastic measurement model for the UAV,

$$\phi_{\mathbf{r}}(t,\tau) = \mathcal{M}(\mathbf{x}(t,\tau) \cdot \mathbf{r}_0 + \mathbf{n}(t,\tau) \tag{8}$$

Here, provided that the UAV might be able to perform multiple rounds of measurements at each discrete time-step t, thus we use another index $\tau \in \{0, 1, ..., \tau^{max}\}$ to denote each round of the measurement. Additionally, $\phi_{\mathbf{r}}(t,\tau) \in \mathbb{R}^{m_t}$ denotes the measurement with its dimension m_t; $\mathcal{M}(\mathbf{x}(t),\tau) \in \mathbb{R}^{\geqslant \approx \times \mathbb{K}}$ is the measurement matrix depending on the UAV's current position $\mathbf{x}(t)$; and $\mathbf{n}(\mathbf{t},\tau) \in \mathbb{R}^{\mathbf{m_t}}$ denotes the measuring noise, assumed to follow a Gaussian distribution (i.i.d.) with zero-mean and variance $V(t,\tau) \in \mathbb{R}^{m_t \times m_t}$.

We shall remark that the measurement matrix $\mathcal{M}(\mathbf{x}(t),\tau)$ is not specified in above model (8). In fact, it can be realized by various means according to the UAV's position $\mathbf{x}(t)$ as well as the sensors equipped on-board. For instance, suppose that a LiDAR senor is equipped with the UAV, which sends laser pulses rotating in a 360-degree range and collects the reflected point cloud within a given radius R. Then, the measurement matrix can be expressed as

$$\mathcal{M}(\mathbf{x}(t),\tau) := [\mathbf{e}_n]^\top_{n \in \mathcal{N}_{\mathbf{x}(t)}} \tag{9}$$

where $\mathbf{e}_n$ denotes the unit vector, i.e., the n-th column of an identity matrix with appropriate dimensions; and the set $\mathcal{N}_{\mathbf{x}(t)}$ contains all indices of the boundaries that fall into the sensing range of the LiDAR. Note that, in this case, the dimension of the produced measurement $\phi_r(t)$ might also be varying over time; thus, we denote it as m_t. An illustration of the specification of measurements is provided in Fig. 4. In this example,

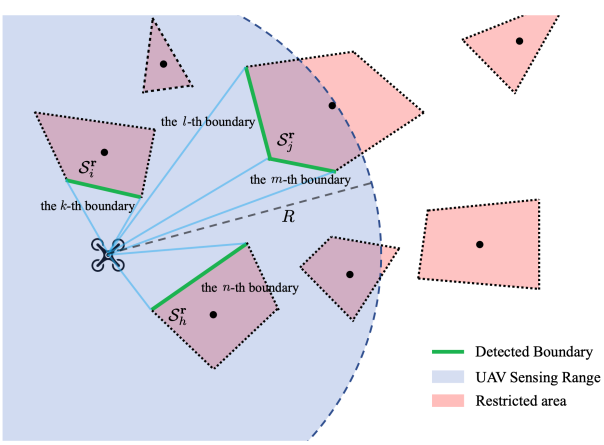

Fig. 4. UAV's measurement on the environment via equipped sensors

the measurement matrix has $\mathcal{M}(\mathbf{x}(t),\tau) = [e_k, e_l, e_m, e_n]^\top$. Indeed, such an UAV-LiDAR framework has been broadly studied in the literature for various applications, including object detection [30], [31], environment mapping [32], [33], etc.

With the help of the measurement model (8), we are now ready to present the RLS estimator for the distance $\mathbf{r}_0$. Suppose that the estimates are generated by the mean $\hat{\mathbf{r}}(t,\tau) \in \mathbb{R}^K$ and covariance $\Sigma(t,\tau) \in \mathbb{R}^{K \times K}$, subsequently the RLS estimator carries out the following recursions:

$$\Sigma(t,\tau+1) = \left(\Sigma(t,\tau)^{-1} + \mathcal{M}(t,\tau)^\top \mathcal{M}(t,\tau)\right)^{-1} \tag{10}$$

$$\begin{aligned}\widehat{\mathbf{r}}(t,\tau+1) =& \widehat{\mathbf{r}}(t,\tau) + \Sigma(t,\tau+1)\mathcal{M}(t,\tau)^\top \left(\phi_{\mathbf{r}}(t,\tau)\right. \\ & \left. - \mathcal{M}(t,\tau)\widehat{\mathbf{r}}(t,\tau)\right)\end{aligned} \tag{11}$$

Note that, at each time-step t, we initialize the mean and variance by $\widehat{\mathbf{r}}(t,0) = \widehat{\mathbf{r}}(t-1,\tau^{\max})$ and $\Sigma(t,0) = \Sigma(t-1,\tau^{\max})$, and the initialization of $\hat{\mathbf{r}}(t,\tau)$ and $\Sigma(0,0)$ will be discussed later; see details in Section III-B.

Furthermore, a few remarks should be also added regarding the RLS estimator (10, 11). First of all, it is worth emphasizing that estimates in (10, 11) are sequentially generated in an online manner, i.e., the UAV moves to a new position, obtains the latest measurements, and updates its estimates on the environment. This is indeed promising for the development of the online trajectory planning algorithm. Second, the above estimator (10, 11), in fact, has the same form of the classical Kalman filtering, in which only the correction step is present since the unknown $\mathbf{r}_0$ is supposed to be stationary. However, due to the sequential process as mentioned before, the obtained measurements are not independent with respect to the time-step t, and thus the standard statistical properties of the Kalman filter no longer hold for the estimates $\hat{\mathbf{r}}(t,\tau)$ and $\Sigma(0,0)$ generated in this case. This actually brings significant challenges for using the produced estimates directly to guide the UAV trajectory planning. Third, to solve the CC-QP problem with the unknown r_i^k's replaced by some certain surrogates, a naive approach is to simply use the mean value $\hat{\mathbf{r}}(t,\tau)$. However, considering that it is unknown (even with probability) if $\hat{\mathbf{r}}(t,\tau)$ is guaranteed to be larger than r_i^k, a critical issue here is the safety of the generated trajectory, i.e., the original chance constraints might end up being violated $\hat{\mathbf{r}}(t,\tau)$ is too small. Therefore, in order to guide the UAV trajectory planning with safety ensured, one has to develop the more sophisticated surrogate of the unknown distances by using the estimates. To do so, we will introduce a new notion of D-UCB in the next subsection.

B. *Construction of D-UCB*

To be clear, let us first recall that, in the original CC-QP (or CC-MIQP) formulation, the UAV is required to bypass all geo-fences with probability at least α. Provided that the value of $\mathbf{r}_0$ basically determines the areas covered by the geo-fences, therefore to ensure the UAV's safety and meanwhile to reduce conservatism, an appropriate surrogate of the unknown $\mathbf{r}_0$, denoted as $\tilde{\mathbf{r}} \in \mathbb{R}^K$, is expected to possess at least the following two features: i) it should be guaranteed with probability at least α that $\tilde{\mathbf{r}}$ is no less than $\mathbf{r}_0$ element-wisely, i.e., $\mathbf{Pr}\left(\tilde{\mathbf{r}} \succeq \mathbf{r}_0\right) \geq \alpha$; and ii) $\tilde{\mathbf{r}}$ should be as small as possible to reduce the conservatism. Toward this end, we construct the key notion of D-UCB $\widetilde{\mathbf{r}}(t) \in \mathbb{R}^K$ which is updated by the mean $\widehat{\mathbf{r}}(t,\tau)$ and variance $\sum(t,\tau)$ as follows,

$$\widetilde{\mathbf{r}}(t) := \widehat{\mathbf{r}}\left(t, \tau^{\max}\right) + \beta(\alpha, t) \cdot \operatorname{diag}^{1/2}\left(\Sigma\left(t, \tau^{\max}\right)\right) \quad (12)$$

Note that the operator $\operatorname{diag}^{1/2}(\cdot) : \mathbb{R}^{K\times K} \to \mathbb{R}^K$ maps the square root of diagonal elements of a matrix to a vector, and the parameter $\beta(\alpha, t)$ depending on the pre-determined threshold α will be specified soon. To confirm that the two above mentioned features are satisfied with the definition of D-UCB (12), we show, in the following proposition, how $\mathbf{r}_0$ is guaranteed to be upper bounded by $\widetilde{\mathbf{r}}(t)$.

Proposition 1: Suppose that the estimates $\widehat{\mathbf{r}}(t, \tau)$ and $\Sigma(t, \tau)$ are generated by the RLS estimator (10, 11) with the initialization $\widehat{\mathbf{r}}(0, 0)$ and $\underline{\sigma} \cdot \mathbf{I} \preceq \Sigma(0, 0) \preceq \bar{\sigma} \cdot \mathbf{I}$, and the variance of the noise $\mathbf{n}(t, \tau)$ is bounded by $V(t, \tau) \preceq \bar{v} \cdot \mathbf{I}$. Let $\tilde{\mathbf{r}}(t)$ be defined as in (12) and the parameter $\beta(\alpha, t)$ be satisfied with

$$\begin{aligned}\beta(\alpha, t) \geq & \sqrt{K} \cdot \|\widehat{\mathbf{r}}(0, 0) - \mathbf{r}_0\| / \sqrt{\underline{\sigma}} \\ & + K\bar{v}^2 \cdot \sqrt{\log\left(\frac{\bar{\sigma}/\underline{\sigma} + \bar{\sigma}t}{(1-\alpha)^{2/K}}\right)}\end{aligned} \quad (13)$$

then it holds that,

$$\Pr\left(\mathbf{r}_0 \preceq \tilde{\mathbf{r}}(t)\right) \geq \alpha, \quad \forall t \in \mathcal{T} \quad (14)$$

By constructing the D-UCB $\widetilde{\mathbf{r}}(t)$ satisfying (14), we can now leverage the realizations of $\tilde{\mathbf{r}}(t)$ to replace the unknown r_i^k 's in the CC-QP or CC-MIQP formulation. As a direct consequence, it is ensured that the produced trajectory has to be feasible with respect to the chance constraint (5). Before the end of this subsection, let us also make a few remarks on the notion of D-UCB as well as the above Proposition 1.

Remark 2: Despite the requirement as stated in Proposition 1 that the selection of parameter $\beta(\alpha, t)$ should be satisfied with the inequality (13), we yet emphasize that one does not need to precisely know the variables involved in the right-hand-side of (13) as prior information. Instead, we are only required to ensure the following two conditions: i) $\beta(\alpha, t)$ increases with respect to the time-step t at the rate of $\mathcal{O}(\sqrt{\log t})$; and ii) it should be guaranteed with the initialized $\beta(\alpha, 0)$ that

$$\widehat{\mathbf{r}}(0, 0) + \beta(\alpha, 0) \cdot \operatorname{diag}^{1/2}(\Sigma(0, 0)) \succeq \mathbf{r}_0 \quad (15)$$

Indeed, the condition (12) assures that D-UCB is a valid upper bound of the true $\mathbf{r}_0$ at the very beginning. It also helps specify the initialization of $\widehat{\mathbf{r}}(0, 0)$ and $\Sigma(0, 0)$

Remark 3: In the computation of D-UCB; see equation (12), $\widetilde{\mathbf{r}}(t)$ is updated at each time-step t when the maximum number of measurements are imposed in the estimates, i.e., $\widehat{\mathbf{r}}\left(t, \tau^{\max}\right)$ and $\Sigma\left(t, \tau^{\max}\right)$. This is primarily due to our belief that the diagonal elements of the covariance matrix $\Sigma(t, \tau)$, in general, will be reduced as more measurements are incorporated; in fact, such a result will be explicitly shown in the next section; see Proposition 2. That is to say, the upper bounds will shrink to the smallest possible values at the time $\tau^{\max}$, which implies that $\widetilde{\mathbf{r}}(t)$, generated by (12), is expected to be less conservative for planning the UAV trajectory. For the same purpose, one can alternatively compute the D-UCB as

$$\widetilde{\mathbf{r}}(t) := \min_{0 \leq \tau \leq \tau^{\max}} \left\{\widehat{\mathbf{r}}(t, \tau) + \beta\left(\alpha^{1/\tau^{\max}}, t\right) \cdot \operatorname{diag}^{1/2}(\Sigma(t, \tau))\right\} \quad (16)$$

where the minimization operator $\min\{\cdot\} : \mathbb{R}^K \to \mathbb{R}^K$ is also performed element-wisely. Notice that the confidence level is enlarged to $\alpha^{1/\tau^{\max}}$ in this case. Additionally, the computation of $\tilde{\mathbf{r}}(t)$ in (16) is valid, due to the fact that, at each time-step t, $\widehat{\mathbf{r}}(t, \tau)$'s are independent with respect to the index τ since the measuring noise $\mathbf{n}(t, \tau)$ is assumed to be $i.i.d.$ generated.

IV. UAV Online Trajectory Planning

A. The Online Planning Algorithm

With the help of the RLS estimator of the unknown distance and then the constructed D-UCB $\tilde{\mathbf{r}}(t)$ as an appropriate surrogate, we are now in the position to present the online trajectory planning algorithm. The basic procedure of the algorithm is as follows. At each time-step t, the UAV receives a sequence of the measurements $\phi_{\mathbf{r}}(t, \tau)$ where $0 \leq \tau \leq \tau^{\max}$, updates D-UCB $\tilde{\mathbf{r}}(t)$ based on the latest estimates $\widehat{\mathbf{r}}(t, \tau)$ and $\Sigma(t, \tau)$, and replans the trajectory according to $\widetilde{\mathbf{r}}(t)$. Then, the UAV is expected to execute only step of the replanned trajectory, and the above process is repeated until reaching the destination. More specifically, the UAV trajectory replanning at each timestep t is performed by solving the following mixed-integer quadratic program (MIQP):

$$\min_{\{\mathbf{u}(s)\}_{s=t}^{T-1}} \sum_{s=t}^{T-1} \|\mathbf{u}(s)\|^2 \quad \text{(MIQP-t)}$$

s.t.

$$\begin{aligned}
& \begin{bmatrix} \mathbf{x}(s+1) \\ \mathbf{v}(s+1) \end{bmatrix} = A \begin{bmatrix} \mathbf{x}(s) \\ \mathbf{v}(s) \end{bmatrix} + B\mathbf{u}(s), t \leq s \leq T-1 \\
& \mathbf{x}(s) = \mathbf{x}(t), \mathbf{v}(s) = \mathbf{v}(t) \\
& \mathbf{x}(T) = \mathbf{x}^{\text{dest}} \\
& \mathbf{x}(s) \in \mathcal{K}, t \leq s \leq T; \\
& \begin{bmatrix} \cos\left(\theta_i^k\right) \\ \sin\left(\theta_i^k\right) \end{bmatrix}^\top (\mathbf{x}(s) - \mathbf{p}_i) + M\left(1 - z_i^k\right) \geq \tilde{r}_i^k(t) \\
& \sum_{k=1}^{K_i} z_i^k \geq 1, z_i^k \in \{0, 1\}, 1 \leq k \leq K_i, i \in \mathcal{I}
\end{aligned}$$

Note that, in the above MIQP $-t$ formulation, the second constraint basically ensures that the UAV's starting position (as well as the velocity) has to be identical with the one where it currently locates. In addition, the surrogate parameter $\tilde{r}_i^k(t)$ is the corresponding component in the vector $\tilde{\mathbf{r}}(t)$ which follows the similar definition as the vector $\mathbf{r}_0$; see Section III-A. It is also worth mentioning that, by applying $\tilde{r}_i^k(t)$'s, the chance constraints have been converted to the deterministic ones in the above MIQP $-t$ and thus the program is expected to be easily handled by commercial solvers, such as CPLEX [34] and GUROBI [35]. To sum up, we outline the complete UAV online planning procedure as Algorithm 1.

Algorithm 1: UAV Online Trajectory Planner

Data: Initialize the UAV's original position $\mathbf{x}(0) = \mathbf{x}^{\text{orig}}$ and velocity $\mathbf{v}(0) = \mathbf{0}$. Set α, $\beta(\alpha, 0)$, $\widehat{\mathbf{r}}(0,0)$, $\Sigma(0,0)$. Determine the planning horizon T and let $t = 0$.

while $t < T$ **do**

- (S.1) Measuring the environment and obtaining the measurements $\boldsymbol{\phi}_{\mathbf{r}}(t, \tau)$;
- (S.2) Performing the RLS estimator in (8) and generating the estimates $\widehat{\mathbf{r}}(t, \tau)$ and $\Sigma(t, \tau)$;
- (S.3) Updating the D-UCB $\widetilde{\mathbf{r}}(t)$ by (9);
- (S.4) Replanning the UAV trajectory by solving the problem (MIQP$-t$);
- (S.5) Executing the input $\mathbf{u}(t)$ and moving to the position $\mathbf{x}(t+1)$ with velocity $\mathbf{v}(t+1)$;
- (S.6) Setting $t \leftarrow t + 1$.

end

B. Performance Analysis

To discuss about the performance of the online planning algorithm, we next evaluate the generated UAV trajectories in the following two aspects: i) feasibility; and ii) (sub-)optimality.

As ensured in Proposition 1 and mentioned in Section III-B, the trajectory produced at each discrete time-step t is immediately guaranteed to satisfy the safety, i.e., feasible with respect to the geo-fence chance constraints, once the condition in (14) holds. Though such a result seems to be quite straightforward, yet we shall highlight that it is actually owing to the main novelty of the notion of D-UCB. In fact, the technique of classical UCB has been widely studied in the context of learning algorithms [27]–[29]. However, a crucial difference here is that, whilst the D-UCB establishes the element-wise upper bound which is defined in the sense of $\mathcal{L}_\infty$ norm, the standard UCB constructs the confidence region in the form of ellipsoid, i.e., in the sense of $\mathcal{L}_2$ norm; see e.g., [36], [37]. To elaborate on this, let us rewrite the vector inequality $\mathbf{r}_0 \preceq \widetilde{\mathbf{r}}(t)$ (by the definition of D-UCB) as follows

$$\|\mathbf{r}_0 - \widehat{\mathbf{r}}\,(t, \tau^{\max})\|_{\mathcal{D}^{-1/2}_{\Sigma(t,\tau^{\max})},\infty} \leq \beta(\alpha, t) \tag{17}$$

where the $\mathcal{L}_\infty$ based norm $\|\cdot\|_{\mathcal{D}_M,\infty} : \mathbb{R}^K \to \mathbb{R}_+$associated with the diagonal matrix of the arbitrary positive definite M, i.e., $\mathcal{D}_M = \text{Diag}\{m_{11}, m_{22}, \cdots, m_{KK}\} \in \mathbb{R}^{K\times K}$, is defined as

$$\|\mathbf{x}\|_{\mathcal{D}_M,\infty} := \max_{1\leq i\leq K} m_{ii} \cdot |x_i| \tag{18}$$

In contrast, the standard UCB establishes the ellipsoid in the form of

$$\|\mathbf{r}_0 - \widehat{\mathbf{r}}\,(t, \tau^{\max})\|_{\Sigma(t,\tau^{\max}),2} \leq \beta'(\alpha, t) \tag{19}$$

where the $\mathcal{L}_2$ based norm $\|\cdot\|_{M,2} : \mathbb{R}^K \to \mathbb{R}_+$associated with the positive definite M is defined as

$$\|\mathbf{x}\|_M := \sqrt{\mathbf{x}^\top M \mathbf{x}} \tag{20}$$

Here we shall remark that, with the upper bound expressed in (19), it is unclear how to construct the valid surrogate for $\mathbf{r}_0$ to ensure the feasibility of trajectories. It is also unknown before the notion of D-UCB if the standard UCB can be transformed into the desired form of $\mathcal{L}_\infty$ norm.

In terms of the optimality, one should note that it actually requires the D-UCB $\tilde{\mathbf{r}}(t)$ to be as small as possible, such that no conservatism will be present in the generated trajectories. Keep this mind, we next show the following proposition.

Proposition 2: Suppose that the estimates $\widehat{\mathbf{r}}(t, \tau)$ and $\Sigma(t, \tau)$ are generated by the RLS estimator (8), and the variance of the noise $\mathbf{n}(t, \tau)$ is bounded by $V(t, \tau) \preceq \bar{v} \cdot \mathbf{I}$. Let the persistent excitation condition hold at each time-step t with respect to the measurement matrix $\mathcal{M}(t, \tau)$, i.e., there exist $\rho > 0$ and $\tau_0 \in \mathbb{N}_+$such that,

$$\sum_{\tau=1}^{\bar{\tau}} \mathcal{M}(t, \tau)^\top \mathcal{M}(t, \tau) \succeq \rho \cdot \mathbf{I}, \quad \forall \bar{\tau} \geq \tau_0 \tag{21}$$

then it holds almost surely that i) $\lim_{\tau\to\infty} \|\Sigma(t, \tau)\| = 0$; and ii) $\lim_{\tau\to\infty} \|\widehat{\mathbf{r}}(t, \tau) - \mathbf{r}_0\| = 0$.

In fact, it is stated in the above Proposition 2 that, under the persistent excitation condition (21), the D-UCB $\widetilde{\mathbf{r}}(t)$ could be arbitrarily close to the true $\mathbf{r}_0$ as the number of measurement $\tau^{\max}$ increases, i.e., $\widetilde{\mathbf{r}}(t) \xrightarrow{\text{a.s.}} \mathbf{r}_0$ as $\tau^{\max} \to \infty$. We would also like to make a few remarks regarding the proposition.

First, it should be noted that the result $\tilde{\mathbf{r}}(t) \xrightarrow{a.s.} \mathbf{r}_0$ is a bit stronger than what is expected by the chance constraints (5). To clarify, let us recall that the chance constraint (5) basically tolerates the violation of safety restriction with probability no larger than $1 - \alpha$. Therefore, while solving the MIQP problem with each $\tilde{r}_i^k(t)$ arbitrarily close to the true r_i^k, then the generated trajectory will be more conservative than the one obtained by solving the CC-QP problem directly. As a consequence, we can only claim that our algorithm produces the sub-optimal trajectories even if the conditions in Proposition 2 are satisfied. However, we would also like to emphasize that this is arguably the best one can achieve under the scenario where some certain environmental information is unknown.

Second, we shall note that the conditions in Proposition 2 are indeed restrictive: i) the persistent excitation condition (21) may not be always satisfied at all time-steps $t \in \mathcal{T}$; and ii) the number of measurements can neither be infinitely large. Nevertheless, by Proposition 2, it is our intention to highlight that the D-UCB $\widetilde{\mathbf{r}}(t)$ tends to decrease as more measurements are incorporated, which in turn helps the planning of the more optimal trajectories. Furthermore, it would be, in fact, quite unfair to compare the UAV trajectory generated by our online algorithm against the globally optimal one. Whereas the latter is normally produced standing in a global view with all environmental information fully known, we are interested in the scenario where some parts of the geo-fence information is unknown and the UAV has only a limited sensing range. In other words, it is highly possible that the globally optimal

trajectory is not even in the solution space that can be reached by the online algorithm. Therefore, in this work the optimality of our algorithm will be mainly evaluated through numerical results.

V. Plan of Future Development

Although this research project is still in the progress, the above section provides only the theoretical approaches of solving the MIQP-t problem and we have not yet complete the simulation for the numerical result. The numerical examples, however, is the next step for us to validate the proposed iterative method. In computer simulation, we plan to test the proposed algorithm under different maps with random obstacles. The main goal is to validate the successful rate α which ideally will help us pushing the upper confidence bound. Second, we would also study how fast would this learning based algorithm preforms.

One problem we found through the simulation was that in the real-world case, the algorithm usually takes seconds to get the new trajectory. While the UAV is maneuvering, the latest trajectory provided by the solver will be outdated since it was based on the environment estimation seconds ago. This, however, is not acceptable for online planning and will be resulting potential failure of the mission. In order to solve this, one possible solution is that use the reinforcement learning method to estimate the calculation time. Then estimate the future UAV position based on its current velocity and the estimated calculation time. Subsequently, the new trajectory should be calculated based on the estimated location in order to avoid the potential failure due to the calculation delay as mentioned earlier.

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